J3A

A1	IN_VIGN_PROGRAM
A2	
A3	TXD_PROGRAM
A4	GND
A5	OEM_A5
A6	GND
A7	IN_CONSTANT_POWER
A8	12V_FROM_MAIN_RELAY
A9	OEM_A9

J3B

B1	OUT_O2_PRE_1_HEATER
B2	OUT_O2_PRE_2_HEATER
B3	CAN_VEHICLE_B_L
B4	CAN_VEHICLE_B_H
B5	
B6	OUT_O2_POST_1_HEATER
B7	O2_PRE_1_VM
B8	O2_POST_2_RETURN
B9	O2_PRE_2_VM
B10	O2_POST_1_RETURN
B11	
B12	OUT_O2_POST_2_HEATER
B13	O2_PRE_1_PUMP_DRIVE
B14	IN_O2_POST_2
B15	O2_PRE_2_PUMP_DRIVE
B16	IN_O2_POST_1
B17	
B18	
B19	O2_PRE_1_PUMP_SENSE
B20	O2_PRE_1_REF_CELL
B21	O2_PRE_2_PUMP_SENSE
B22	O2_PRE_2_REF_CELL
B23	OUT_MAIN_RELAY
B24	

J3C

C1	IN_MAF
C2	IN_MAP
C3	IN_IAT
C4	5V_MAP_MAF
C5	IN_OIL_LEVEL
C6	OUT_INJ4
C7	OUT_INJ3
C8	
C9	OUT_VVT_EXHAUST
C10	OUT_VVT_INTAKE
C11	
C12	OUT_THERMOSTAT
C13	OUT_INJ1
C14	GNDA_C14
C15	
C16	
C17	
C18	OUT_VALVETRONIC_EMERGENCY
C19	BSD_ALTERNATOR
C20	
C21	OUT_EVAP
C22	
C23	
C24	
C25	OUT_INJ2
C26	
C27	IN_CRANK
C28	IN_CLT
C29	IN_CAM_INTAKE
C30	IN_CAM_EXHAUST
C31	IN_TPS1
C32	IN_TPS2
C33	KNOCK_12_A
C34	KNOCK_34_A
C35	GNDA_C35
C36	GNDA_C36
C37	GNDA_C37
C38	CAN2_H
C39	
C40	DISA_MOTOR_1
C41	DISA_MOTOR_2
C42	OUT_ETB_NEG
C43	OUT_ETB_POS
C44	
C45	IN_DISA
C46	KNOCK_12_B
C47	KNOCK_34_B
C48	GNDA_C48
C49	
C50	5V_TPS
C51	CAN2_L
C52	GNDA_C52

J3D

D1	
D2	
D3	OUT_SAI_RELAY
D4	OUT_FAN
D5	GND
D6	IN_START
D7	GNDA_D7
D8	IN_PPS1
D9	5V_PPS1
D10	OUT_FUEL_PUMP
D11	
D12	GNDA_D12
D13	IN_PPS2
D14	5V_PPS2
D15	
D16	
D17	OUT_TACH
D18	
D19	
D20	
D21	
D22	IN_VSS
D23	IN_CLUTCH
D24	IN_BRAKE_2
D25	
D26	IN_VIGN
D27	MFL_SUPPLY
D28	IN_BRAKE
D29	OUT_AC_RELAY
D30	
D31	
D32	K_LINE
D33	EWS
D34	
D35	OEM_D35
D36	CAN_H
D37	CAN_L
D38	GNDA_D38
D39	IN_RADIATOR_CLT
D40	

J3E

E1	OUT_HV_IGN_4
E2	OUT_HV_IGN_1
E3	
E4	GND
E5	OUT_HV_IGN_2
E6	
E7	
E8	
E9	OUT_HV_IGN_3

J1A

IN_IGN_1	1A
GND	2A
IN_IGN_2	3A
IN_IGN_3	4A
IN_MAP	5A
IN_START	6A
5V_PPS2	7A
IN_PPS2	8A
OUT_ETB_NEG	9A
IN_IGN_4	10A
OUT_FUEL_PUMP	11A
IN_BRAKE	12A
CAN_L	13A
5V_MAP_MAF	14A
IN_OIL_LEVEL	15A
OUT_TACH	16A
IN_PPS1	17A
GNDA_C35	18A
CAN_H	19A
EWS	20A
IN_VIGN	21A
IN_BRAKE_2	22A
OUT_SAI_RELAY	23A
IN_RADIATOR_CLT	24A
OUT_FAN	25A
KNOCK_12_A	26A
K_LINE	27A
GNDA_D38	28A
OUT_AC_RELAY	29A
IN_CLUTCH	30A
OUT_INJ4	31A
GNDA_D12	32A
5V_PPS1	33A
GNDA_D7	34A

J1B

IN_VSS	1B
KNOCK_12_B	2B
OUT_VVT_INTAKE	3B
OUT_VVT_EXHAUST	4B
OUT_INJ2	5B
BSD_ALTERNATOR	6B
OUT_INJ3	7B
OUT_VALVETRONIC_EMERGENCY	8B
CAN2_H	9B
GNDA_C52	10B
CAN2_L	11B
IN_TPS2	12B
KNOCK_34_B	13B
IN_IAT	14B
KNOCK_34_A	15B
5V_TPS	16B
GNDA_C37	17B
IN_CAM_EXHAUST	18B
GNDA_C48	19B
OUT_EVAP	20B
IN_TPS1	21B
OUT_INJ1	22B
IN_CAM_INTAKE	23B
OUT_THERMOSTAT	24B
GNDA_C36	25B
OUT_ETB_POS	26B

J2

OUT_O2_PRE_2_HEATER	1C
DISA_MOTOR_2	2C
O2_POST_1_RETURN	3C
GND	4C
OUT_O2_PRE_1_HEATER	5C
DISA_MOTOR_1	6C
OUT_MAIN_RELAY	7C
IN_CLT	8C
O2_PRE_1_VM	9C
O2_PRE_2_PUMP_SENSE	10C
IN_MAF	11C
O2_PRE_2_REF_CELL	12C
O2_POST_2_RETURN	13C
O2_PRE_1_REF_CELL	14C
GNDA_C14	15C
O2_PRE_1_PUMP_DRIVE	16C
O2_PRE_2_VM	17C
O2_PRE_2_PUMP_DRIVE	18C
IN_O2_POST_2	19C
OUT_O2_POST_2_HEATER	20C
IN_O2_POST_1	21C
O2_PRE_1_PUMP_SENSE	22C
OUT_O2_POST_1_HEATER	23C
IN_CRANK	24C
12V_FROM_MAIN_RELAY	25C
IN_CONSTANT_POWER	26C

IN_VIGN_PROGRAM — PA1

TXD_PROGRAM — PA3

OEM_A5 — PA5

OEM_A9 — PA9

CAN_VEHICLE_B_L — PB3

CAN_VEHICLE_B_H — PB4

IN_DISA — PC45

MFL_SUPPLY — PD27

OEM_D35 — PD35

R9 (DNP)
IN_IGN_1 — OUT_HV_IGN_1

R10 (DNP)
IN_IGN_2 — OUT_HV_IGN_2

R11 (DNP)
IN_IGN_3 — OUT_HV_IGN_3

R12 (DNP)
IN_IGN_4 — OUT_HV_IGN_4

IN_IGN_1 — R1 22 — GATE_1 — Q1 — OUT_HV_IGN_1
Q1: ISL9V3040D3ST
IN_IGN_1 — R5 100k — GND

IN_IGN_2 — R2 22 — GATE_2 — Q2 — OUT_HV_IGN_2
Q2: ISL9V3040D3ST
IN_IGN_2 — R6 100k — GND

IN_IGN_3 — R3 22 — GATE_3 — Q3 — OUT_HV_IGN_3
Q3: ISL9V3040D3ST
IN_IGN_3 — R7 100k — GND

IN_IGN_4 — R4 22 — GATE_4 — Q4 — OUT_HV_IGN_4
Q4: ISL9V3040D3ST
IN_IGN_4 — R8 100k — GND

GND

rusEFI
Sheet: /
File: bmw-e46-tomas.kicad_sch
Title: BMW E46 N42 134 to Superseal 60+26
Size: A2 | Date: 2026-10-21 | Rev: a
KiCad E.D.A. 10.0.6 | Id: 1/1